

Question number	Answer	Additional guidance	Mark
1(a)(i)	B phosphorylation of hexoses		Computer (1)

Question number	Answer	Additional guidance	Mark
1(a)(ii)	C increases, causing a decrease in blood pH		Computer (1)

Question number	Answer	Additional guidance	Mark
1(b)(i)	A as molecules containing 2 carbon atoms produced by the link reaction		Computer (1)

Question number	Answer	Additional guidance	Mark
1(b)(ii)	An answer that includes the following points: - double membrane structure with cristae (1) (mitochondrial) matrix identified as location of Krebs cycle reactions (1)	ALLOW Krebs cycle if arrow points to correct location. Allow without arrow labelled	Graduate (2)

Question number	Answer	Additional guidance	Mark
1(b)(iii)	A description that includes five of the following points: - hydrogen atoms are transported to the electron transport chain (1) - by (the coenzymes) NAD and FAD (1) - electrons pass along the electron transport chain releasing energy (1) - that is used to move protons to the intermembrane space (1) - protons diffuse (back into the mitochondrial matrix) through ATP synthase (1) (catalysing) the formation of ATP from ADP and Pi (1)	ALLOW hydrogen ions and electrons ALLOW reduced NAD/NADH/NADH₂ and reduced FAD/FADH/ FADH₂ ALLOW H⁺ /Hydrogen ions IGNORE ATPase ALLOW Phosphorylating ADP ALLOW correct equation	Expert (5)

